

Sitting and endothelial dysfunction: The role of shear stress



Summary Sedentary activity is a modifiable life-style behavior and a key component in the etiology of atherosclerotic cardiovascular disease (ACVD). US adults and children spend more than half their waking time in sedentary pursuits. Sedentary activity has been shown to result in impaired insulin sensitivity, impaired metabolic function and attenuated endothelial function, which are classic markers of ACVD. Sedentary activity is defined as ‘sitting without otherwise being active.’ This behavior promotes reduced muscular activity of the lower extremities which decreases leg blood flow, increases blood pooling in the calf, augments mean arterial pressure, and deforms arterial segments resulting in low mean shear stress (SS). SS activates distinct physiological mechanisms which have been proposed to be protective against ACVD; specifically through a SS-induced endothelium-derived nitric oxide mechanism. Reduced bioavailability of nitric oxide creates a pro-oxidant milieu resulting in increased oxidative stress. There is sufficient evidence which demonstrates that endothelial function is attenuated in the presence of oxidative stress. Sedentary activity results in low SS in the lower extremities which may result in increased oxidative stress and impaired endothelial function. This review furthers the use of sitting as model to study the effects of inactivity, discusses possible physiological mechanisms and suggests future directions.